13

Electromagnetic Induction

13.1

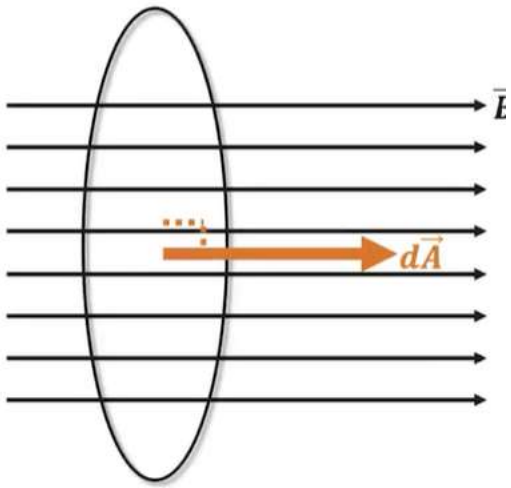
Magnetic Flux

13.1.1

Conceptual Magnetic Flux and Vector Fields

FLUX

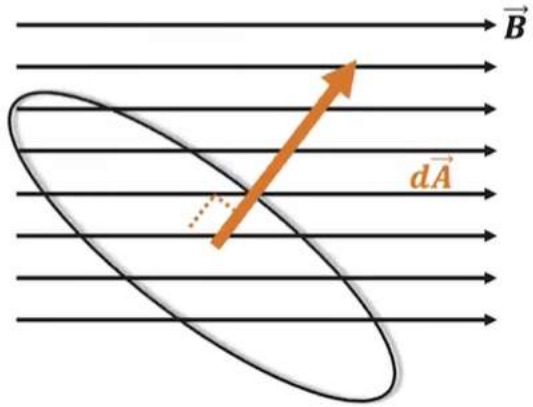
- Flux is an amount of stuff going through an area
- We've seen electric flux through a surface area: $\oint \vec{E} \cdot d\vec{A}$
- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $\mathrm{d}\vec{A}$ vector points **perpendicular** to the area of the loop



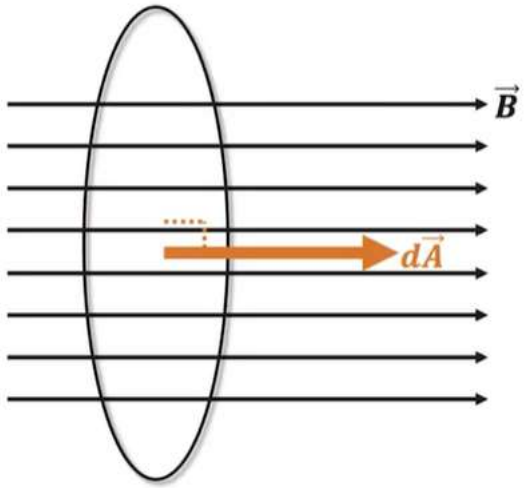
FLUX

Remember dot product!

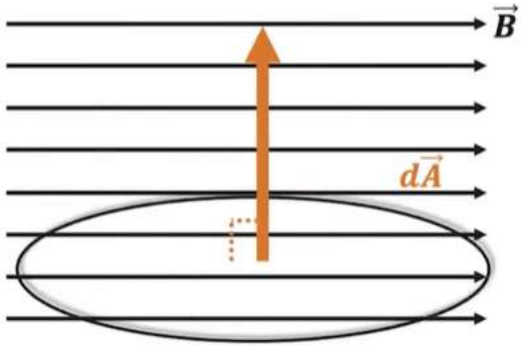
$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$



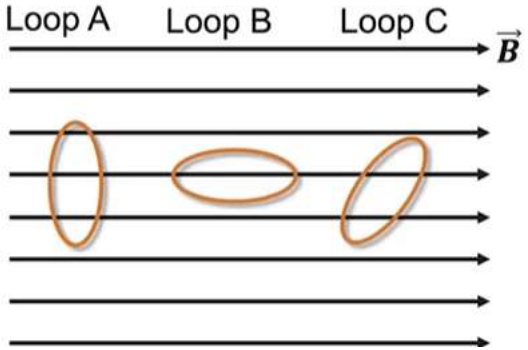
If $d\vec{A} \parallel \vec{B}$, then $\vec{B} \cdot d\vec{A} = |\vec{B}||d\vec{A}|$



If $d\vec{A} \perp \vec{B}$, then $\vec{B} \cdot d\vec{A} = 0$

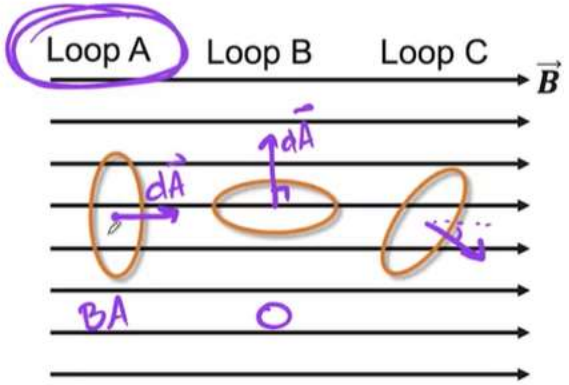


Which loop has greater magnetic flux?



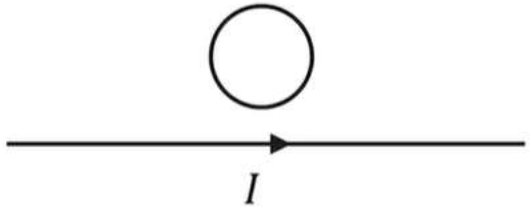
Which loop has greater magnetic flux?

$$\phi_B = \int \underbrace{\vec{B} \cdot d\vec{A}}_{B\,dA\cos\theta}$$



A small, circular loop of conducting wire is near a long, straight wire that carries a steady electric current I to the right, as shown below.

Which of the following actions will result in an increase in magnetic flux through the wire loop?



1. The wire loop is pulled vertically upward in the plane of the page (away from the wire)
2. The current in the straight wire is steadily increasing.
3. The wire loop is rotated about a diameter through a small angle so that it is no longer in the plane of the page.

A. 1 only

B. 2 only

C. 3 only

D. Either 1 or 2

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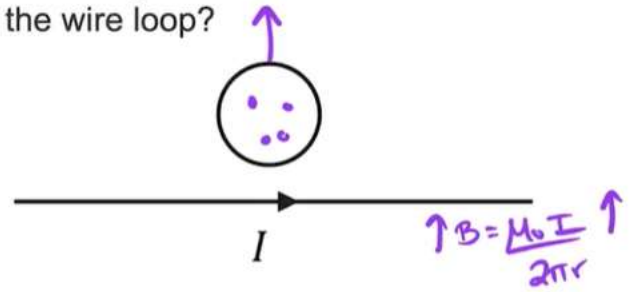

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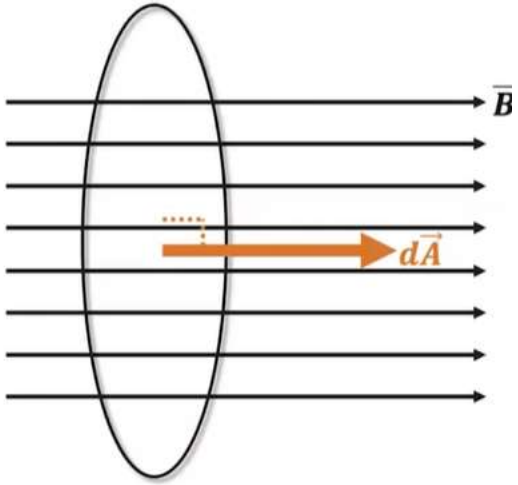
What is the flux through a hoop of radius R?



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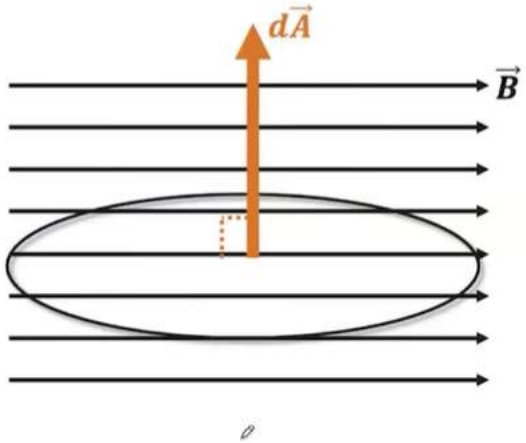

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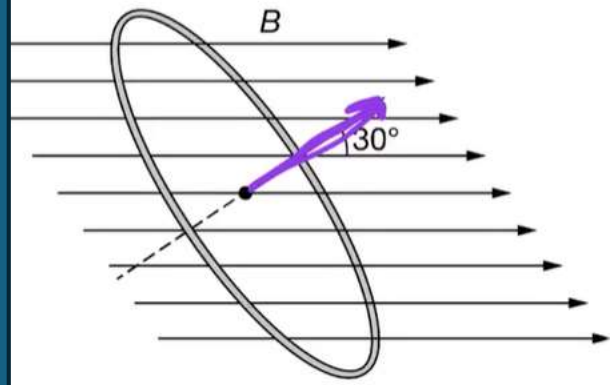


Takeaways

Magnetic Flux depends on:

- Strength of Magnetic Field
- Size of loop area exposed to field
- Angle of loop in relation to field

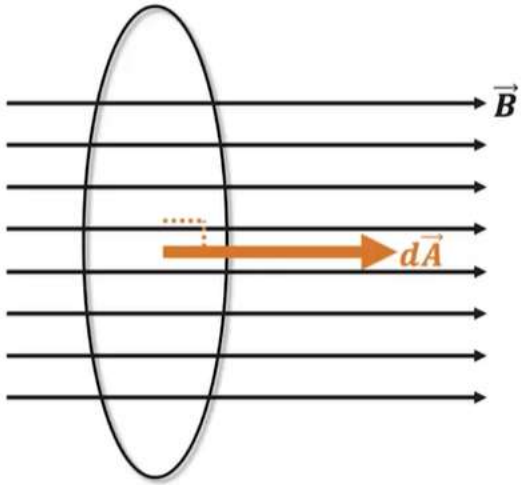
13.1.2

Calculating Common Situations of Magnetic Flux

Integrate or Not?

FLUX

- Flux is an amount of stuff going through an area
- We've seen electric flux through a surface area: $\oint \vec{E} \cdot d\vec{A}$
- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $d\vec{A}$ vector points **perpendicular** to the area of the loop



FLUX

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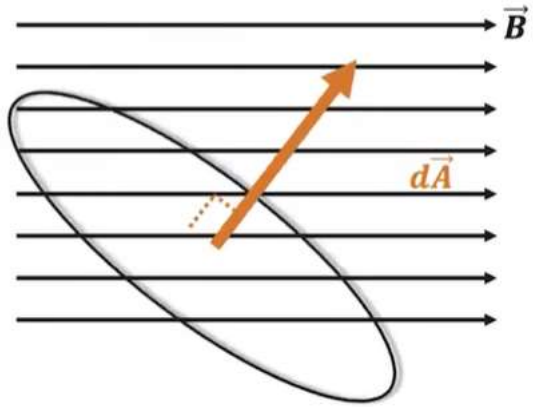

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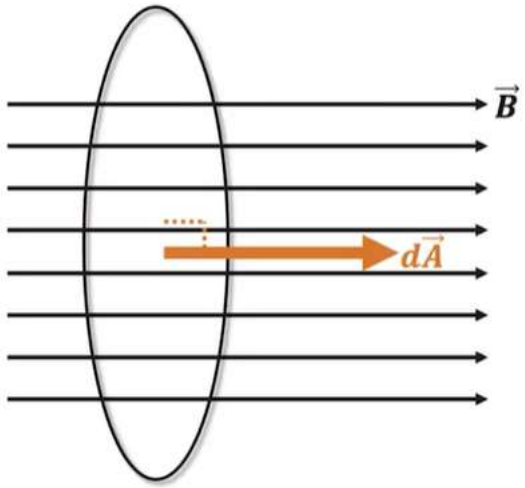

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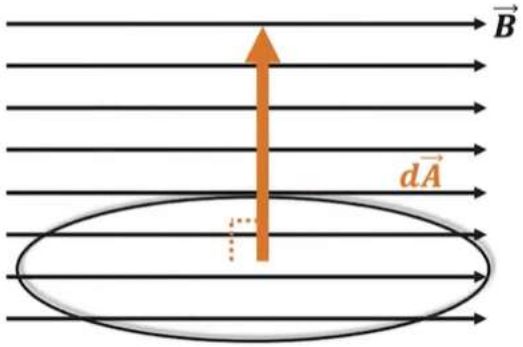
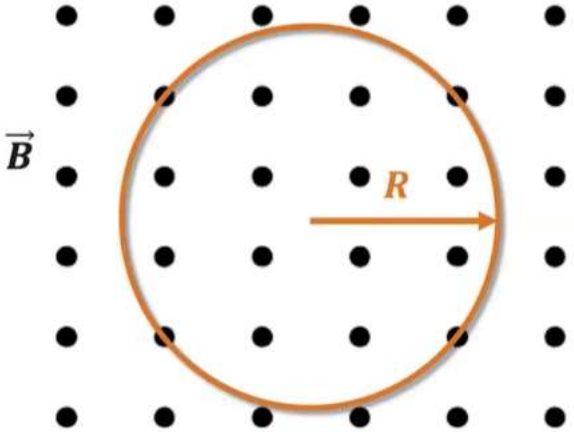


Magnetic flux with changing area

A circular current-carrying loop lies so that the plane of the loop is perpendicular to a constant magnetic field of strength $\vec{B}$.

Suppose that the radius R of the loop could be made to increase with time t so that $R = at$, where a is a constant.

What is the magnetic flux through the loop?



Source: Released AP Exam. Taken (and modified) from AP Classroom

Magnetic flux with changing area

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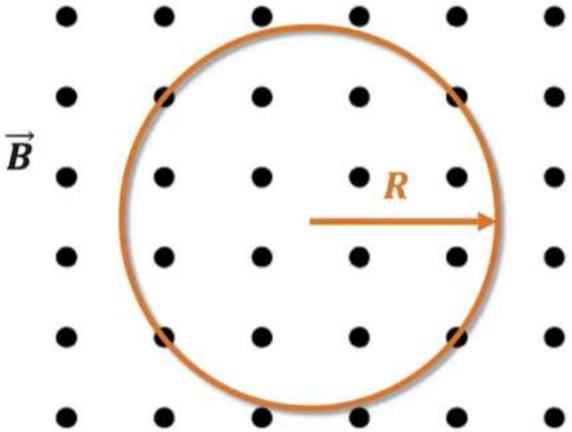

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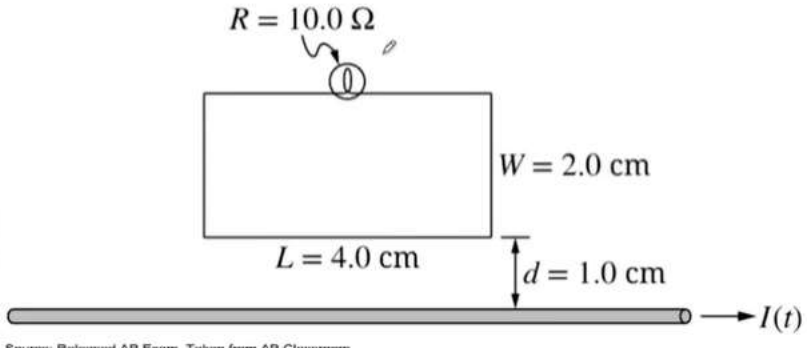
Loop and Wire FRQ

A lightbulb of **resistance $R = 10\ \Omega$** is connected to a rectangular loop of wire of negligible resistance near a very long current-carrying wire.

The rectangular loop has **length $L = 4.0\ \text{cm}$** and a **width of $w = 2.0\ \text{cm}$** and is positioned so one of the longer sides of the loop is a **distance $d = 1.0\ \text{cm}$** above and parallel to the long wire, as shown.

The current in the long wire is initially flowing to the right and is given by $I = C - Dt$, where $C = 10.0\ A$ and $D = 2.0\ A/s$.

At time $t = 5.0\ s$, the current in the long wire is instantaneously zero as the current changes direction.



Loop and Wire FRQ

Calculate the magnetic flux through the loop due to only the long wire at time $t = 3.0s$.

$I = C - Dt$

$B = \frac{\mu_0 I}{2\pi r}$

$\Phi_B = \int \underbrace{\vec{B} \cdot d\vec{A}}$ $\quad \vec{B} \parallel d\vec{A}$

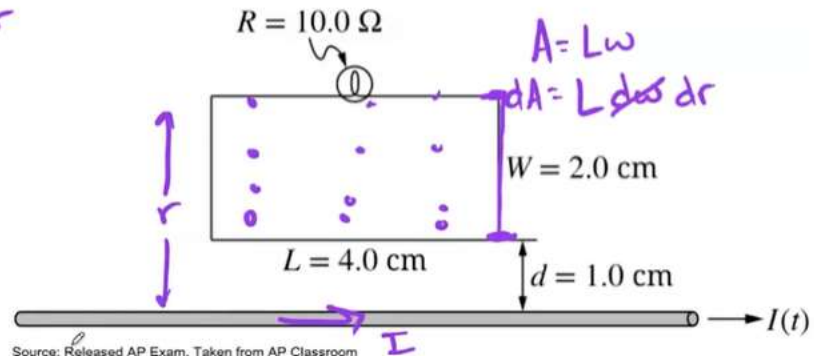
$= \int B\,dA = \int \frac{\mu_0 I}{2\pi r} L\,dr$

$= \frac{\mu_0 I L}{2\pi} \int_d^{d+w} \frac{1}{r}\,dr$

$= \frac{\mu_0 I L}{2\pi} \ln|r| \Big|_d^{d+w}$

$A = Lw$

$dA = L\,\cancel{dw}\,dr$



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Loop and Wire FRQ

Calculate the magnetic flux through the loop due to only the long wire at time $t = 3.0s$.

$I = C - Dt$

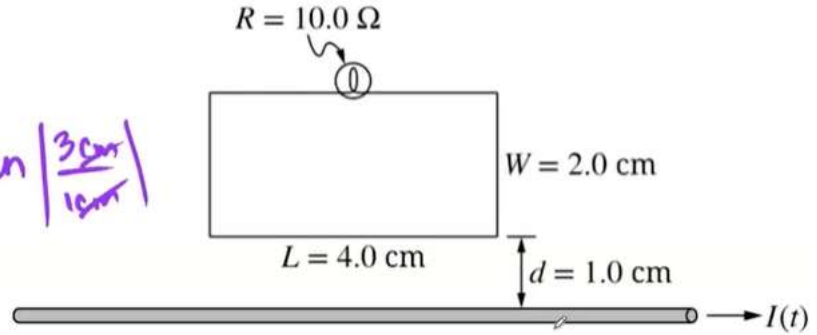
$10 - 2(3) = 4A$

$\Phi_B = \frac{\mu_0 I L}{2\pi} \ln|r| \Big|_d^{d+w}$

$\boxed{\Phi_B = \frac{\mu_0 I L}{2\pi} \ln\left|\frac{d+w}{d}\right|}$

$= \frac{(4\pi \times 10^{-7})(4A)(.04m)}{2\pi} \ln\left|\frac{3cm}{1cm}\right|$

$= \boxed{3.52 \times 10^{-8}\ T \cdot m^2}$



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Takeaways

When to integrate for flux:

You only need to integrate if the magnetic field isn't constant over the entire area (it depends on r, or width, etc)

13.1

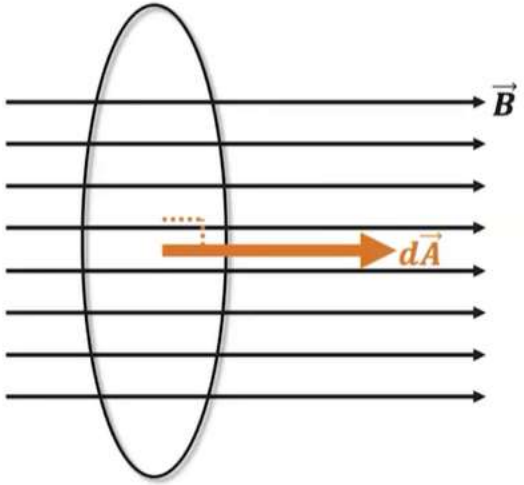
Electromagnetic Induction

13.2.1

Faraday’s Law Component Examples

MAGNETIC FLUX

- Now we have magnetic flux through a loop area: $\int \vec{B} \cdot d\vec{A}$
- The $d\vec{A}$ vector points **perpendicular** to the area of the loop
- Two ways to change Magnetic Flux: Change the field, or change the area exposed to the field (size or orientation/angle)



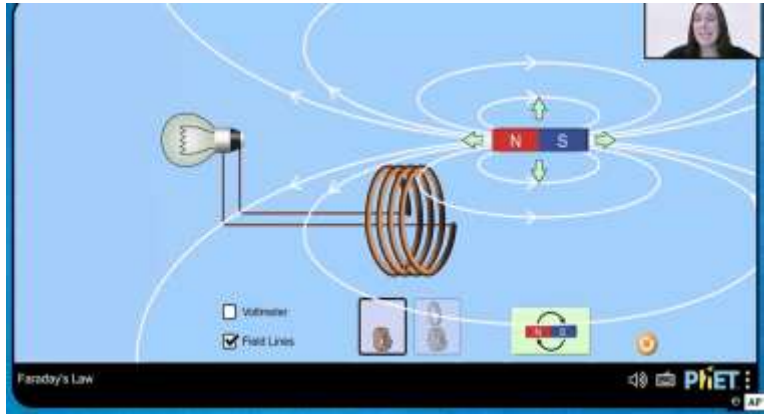
Ohm's Law:

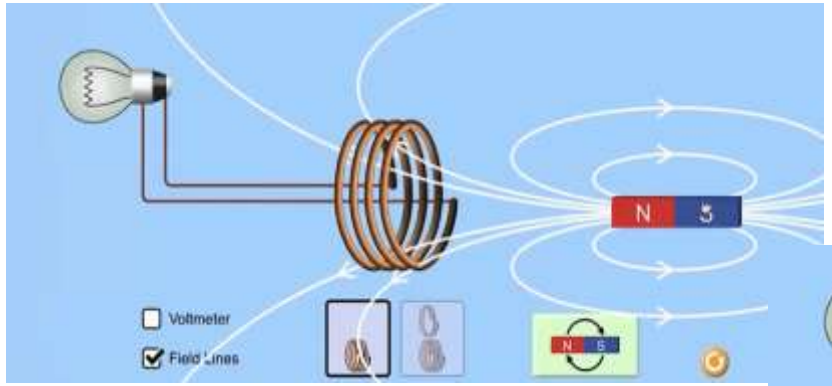
$$\mathbf{I} = \frac{\Delta \boldsymbol{V}}{\boldsymbol{R}}$$

Faraday's Law:

A changing magnetic flux induces an emf in the system

$$emf = -\frac{d\Phi_B}{dt}$$


N
S
Voltmeter
Field Lines
Faraday's Law
PhET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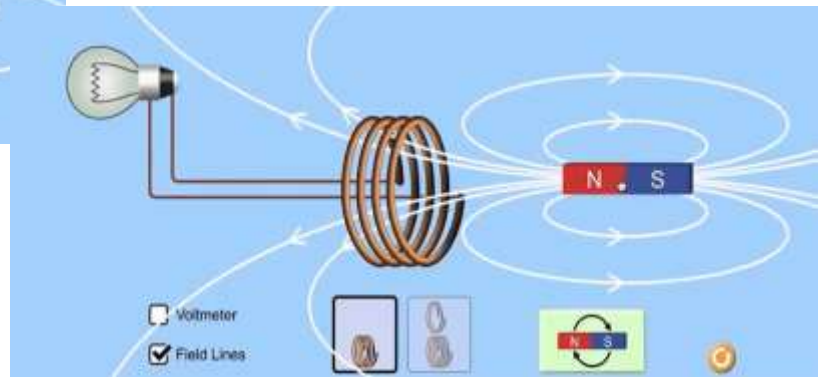
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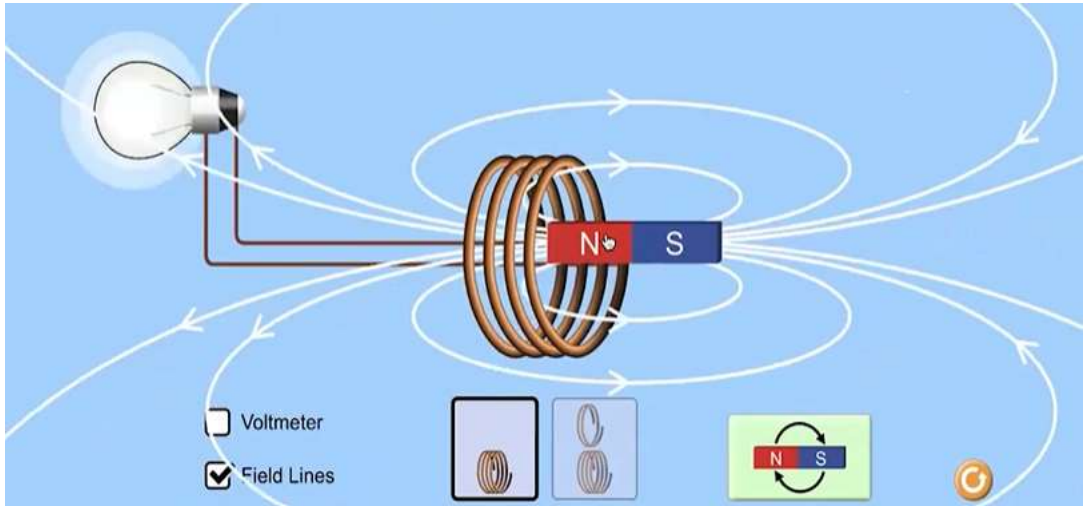
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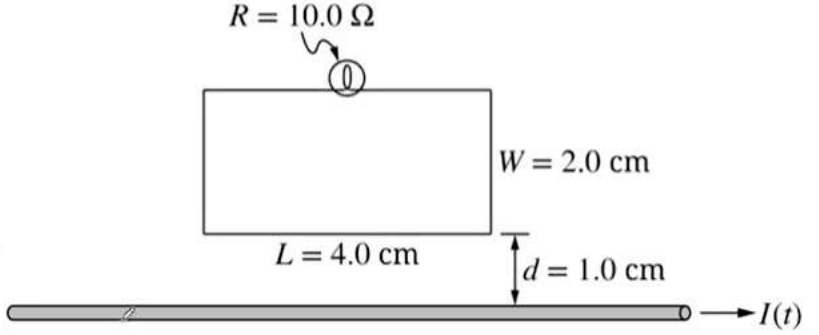
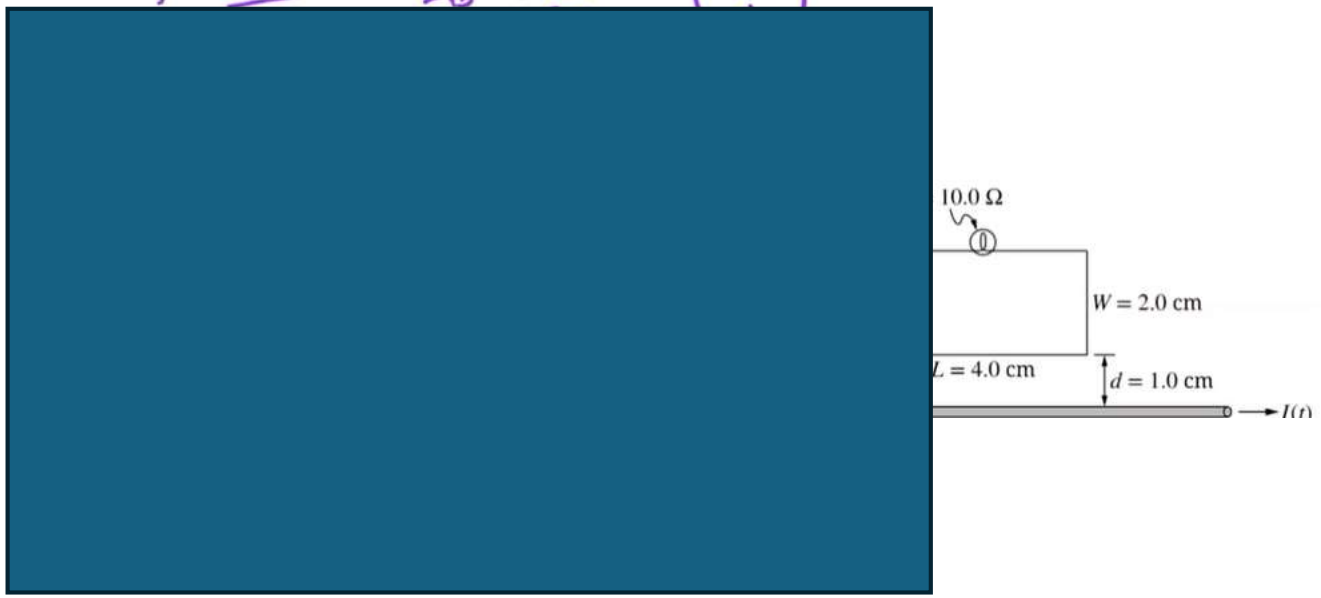


Source: Released AP Exam. Taken from AP Classroom

Loop and Wire FRQ

Calculate the magnitude of the induced emf generated around the loop at time $t = 3.0s$.

$emf = -\frac{d\Phi_B}{}$ $\Phi_B = \frac{\mu_0 I L}{} \ln\left(\frac{d+w}{}\right)$ $I = (C - Dt)$



A circular current-carrying loop lies so that the plane of the loop is perpendicular to a constant magnetic field of strength B. Suppose that the radius R of the loop could be made to increase with time t so that $R = at$, where a is a constant.

What is the magnitude of the emf that would be generated around the loop as a function of t?

$\Phi_B = B\pi a^2 t^2$

A $2\pi B a^2 t$

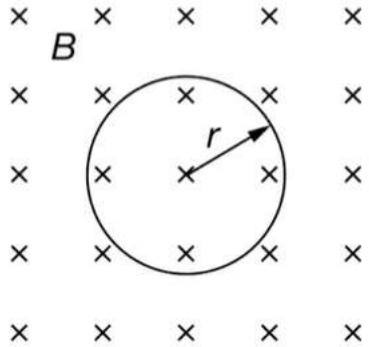
B $2\pi B a t$

C $2\pi B t$

D $\pi B a^2 t$

A single loop of wire is at rest in a magnetic field that is directed into the page, as shown in the figure below. The loop has a radius of r and a resistance of R. The magnetic field has a magnitude B that changes as a function of time t according to the equation $B = \alpha t + \beta t^2$, where α and β are both positive constants in units of T/s and T/s^2, respectively.

Which of the following expressions gives the current in the loop as a function of time t?



Source: Released AP Exam. Taken from AP Classroom

A $\dfrac{\beta t \pi r^2}{R}$

B $\dfrac{(\alpha + 2\beta t)\pi r^2}{R}$

C $\dfrac{(\alpha + \beta t) t \pi r^2}{R}$

D $\dfrac{(\alpha + 2\beta t) 2\pi r}{R}$

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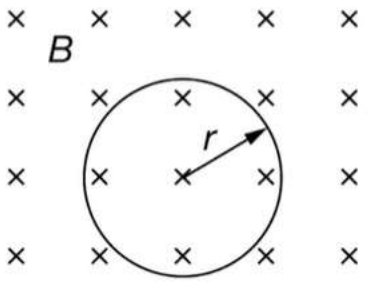
$$\text{emf} = -\frac{d\Phi_B}{dt}$$

$$\Phi_B = \left(\int \vec{B} \cdot d\vec{A}\right) = BA$$

$$= (\alpha t + \beta t^2)\pi r^2$$



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Which of the following expressions gives the current in the loop as a function of time t?

$\Phi_B = (\alpha t + \beta t^2)\pi r^2$

$emf = -\frac{d\Phi_B}{dt} = -(\alpha + 2\beta t)\pi r^2$

$I = \frac{\Delta V}{R} = \boxed{\frac{(\alpha + 2\beta t)\pi r^2}{R}}$

A $\frac{\beta t \pi r^2}{R}$

B $\frac{(\alpha + 2\beta t)\pi r^2}{R}$

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Takeaways

A changing magnetic flux induces an emf in the system

Change the area (size or orientation/angle)

Change the field